Graph-Based Movie Recommendations with Personalized PageRank

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Abstract

This work compares graph-based movie retrieval with popularity and matrix factorization on MovieLens 100K under one shared warm-user protocol. A TensorFlow.js browser application constructs a positive-interaction bipartite graph, implements dangling-aware PageRank and personalized PageRank (PPR), and selects the restart control on validation data. On 861 test users, PPR at alpha = 0.50 achieves Recall@10 of 0.1122, compared with 0.0760 for popularity, 0.0797 for global PageRank and 0.0547 for the repaired MF comparator. The offline result supports testing PPR as the main recommendation strategy, while differing validation objectives, popularity concentration and unavailable historical MF weights limit the deployment claim.

1. Introduction

The task is to retrieve unseen movies from users' positive rating histories. A rating matrix is sparse, and an unobserved rating is missing information rather than a zero-valued preference. A graph can connect a user to movies supported by paths through other users without fitting a rating predictor.

The practical motivation is to determine whether a more personalized retrieval method earns its additional computation. A shared candidate pool and test cohort make the comparison more interpretable than comparing unrelated rating and ranking metrics. For example, user 1 and the training ratings enter the application; an eligible list containing Fargo (ID 100) and Contact (ID 258), with genres and scores, is returned.

Contributions. (1) Implement and verify tensor-based propagation, filtering and shared-cohort evaluation. (2) Compare four methods and diagnose a real MF regularization failure rather than relying on the self-check count alone.

2. Related Work

Haveliwala [1, section 2] explains how a nonuniform personalization vector changes PageRank during propagation. Here that mechanism is adapted to a user-movie graph with a one-hot user restart; the topic-classification component of that web-search paper is not implemented. Its alpha denotes restart probability, whereas this assignment uses alpha for link following.

Uniform-restart PageRank is the structural alternative; popularity is the inexpensive degree-based alternative. MF is the previous assignment's latent-factor comparator, which predicts ratings rather than walk probabilities. These alternatives are retained to test whether personalization adds retrieval value rather than assuming it does.

MovieLens is documented by Harper and Konstan [2] and distributed by GroupLens [3]. Its ratings and timestamps support this offline exercise but do not measure current-session intent or viewing completion. TensorFlow.js 4.22.0 documentation [4] supplies the actual matrix and tensor-lifecycle API contracts used here.

3. Method

Data and graph. Preserve the original week5/u.data and u.item. Within each user, sort by timestamp (movie ID breaks ties), then take floor(0.8n) training and floor(0.1n) validation records, leaving the rest for test. Only training ratings at least 4 create bidirectional user-movie edges. Training-observed IDs map to contiguous indices with no padding node. A missing positive edge is not a dislike.

Propagation. Let Q contain row-normalized transitions for nonzero-degree nodes and zero rows otherwise; d is the current probability on dangling nodes. Starting from p = v, compute **p_next = alpha (Q^T p + d v) + (1 - alpha) v**. Global PageRank uses uniform v; PPR uses a one-hot target-user v. Stop at L1 difference below 0.000001 or 500 iterations; reaching the cap alone is not convergence. Batched tf.matMul operates on the two bipartite blocks, with async reads/yields, retained graph tensors and explicit disposal. Async callbacks are not placed inside tf.tidy [4].

Retrieval and selection. Rank only the 1,611 training-observed movies. Validation excludes all training-rated items; test excludes all training- and validation-rated items, regardless of rating. Sort by descending score and ascending movie ID on ties. Relevance means rating at least 4 in the evaluation split, eligible and in the training pool. Macro Recall@10 averages hit/relevant over the common evaluable users: a positive training edge, MF-scoreable candidates and at least one eligible relevant item. Select one PPR alpha using validation Recall; exact ties choose the smaller alpha. The same alpha controls the primary global PageRank comparison, which is not independently optimized.

MF provenance and checkpoint. The fixed comparator uses 16 factors, Adam lr 0.01, regularization 0.05 divided by actual batch size, batch 512,30 epochs and seed 20261013. Its score is the global training mean plus a latent-factor dot product. The best validation-RMSE factors are restored: epoch 6, validation RMSE 0.9878. No test metric selects a checkpoint. The previous HW5 weights were not persisted: this is a retrained comparator, not the identical saved HW5 model. HW5 used batch 64 / 20 epochs, unseeded initialization and clipped validation predictions; HW6 retains its provided seeded configuration and unclipped validation RMSE. The repaired L2 normalization matches the HW5 convention. These differences qualify the continuity required by assignment section 6; they are disclosed rather than described as exact model reuse.

Environment and reproduction. Windows, browser backend WebGL; TensorFlow.js 4.22.0, D3 7.9.0. GPU/driver identity was not captured, so no device-normalized speed claim is made. Serve the repository root with `python -m http.server 8000`, open /week6/, Build graph, Evaluate, Run tests, then export. Run two-user demo for the controlled lists. The current script SHA256 begins 45d79c3c2da045bc; full hashes are in the audit record. Student repository: a06-personalized-pagerank/week6.

4. Experiments

Quantity	Observed value
Raw data / split	943 users;1,682 metadata movies;100,000 ratings 79,619 train /9,596 validation /10,785 test
Training graph	943 users +1,611 movies =2,554 nodes;45,602 positive interactions
Zero degree / sparsity	1 user;253 movies; displayed full rating-matrix sparsity93.71%
Evaluation cohorts / exclusions	846 validation users;861 test users;13 unseen relevant test movie IDs;82 users without eligible relevant test items

The split is chronological per user, not globally time-consistent. MovieLens timestamps record rating entry, not consumption [2]. The unseen count is distinct movie IDs, not interactions; exclusions must not be added as disjoint groups. This warm-user experiment does not establish cold-start performance.

Verification covers the directed reference graph (uniform restart: C > A > B > D; restart at A: A > C > B > D), normalized rows, valid restarts, mass conservation, nonnegative scores, stationarity, a separate dangling fixture, matrix orientation, recommendation filtering, mappings and public interfaces. The exported check details identify the tested fixtures; toy checks alone do not validate the full real-data comparison.

4. Experiments (continued)

Method	alpha	Test Recall@10	Mean training popularity
popularity	—	0.0760	272.00
pagerank	0.50	0.0797	257.82
ppr	0.50	0.1122	244.21
mf	—	0.0547	153.34

All four rows use 861 users. Validation PPR Recall@10 is 0.1145/0.1048/0.0900/0.0738 for alpha 0.50/0.70/0.85/0.95 respectively; 0.50 is selected before test evaluation. MF uses validation RMSE, a different objective from PPR's validation Recall. MF remains below popularity after its repair; this negative result is retained.

The completed WebGL evaluation took 154.1 s. Global PageRank at 0.50 took 19 iterations with final L1 residual 0.0000005436; the displayed user 2 PPR trace took 21 with residual 0.0000009788. Both satisfy the tolerance; these two traces do not certify every user's convergence. The machine-readable run is run-1-1, exported 20 October 2026. The self-check export is 12 PASS / 0 FAIL / 0 PENDING.

4.1 Two-user evidence and controlled changes

Users 1 and2 are fixed throughout. Their primary-list overlaps are: popularity 1, PageRank 1, PPR 1, MF 5 common movies. The respective list mean popularities (user 1/user 2) are 246.8/262.1,239.4/247.4,228.4/226.7 and138.5/140.9. These are descriptive differences, not a measure of personalization quality. All 100 exported recommendation rows form 10 complete Top-10 lists; no inspected list contains a zero score or a previously rated test-excluded item.

Rank	User1 PPR, alpha0.50	User2 PPR, alpha0.50
1	100 Fargo (1996) Crime / Drama / Thriller; pop 377	181 Return of the Jedi (1983) Action / Adventure / Romance / Sci-Fi / War; pop 339
2	258 Contact (1997) Drama / Sci-Fi; pop 307	9 Dead Man Walking (1995) Drama; pop 182
3	286 English Patient, The (1996) Drama / Romance / War; pop 279	7 Twelve Monkeys (1995) Drama / Sci-Fi; pop 234
4	318 Schindler's List (1993) Drama / War; pop 224	117 Rock, The (1996) Action / Adventure / Thriller; pop 215
5	357 One Flew Over the Cuckoo's Nest (1975) Drama; pop 202	98 Silence of the Lambs, The (1991) Drama / Thriller; pop 308
6	483 Casablanca (1942) Drama / Romance / War; pop 197	121 Independence Day (ID4) (1996) Action / Sci-Fi / War; pop 207
7	222 Star Trek: First Contact (1996) Action / Adventure / Sci-Fi; pop 189	174 Raiders of the Lost Ark (1981) Action / Adventure; pop 312
8	475 Trainspotting (1996) Drama; pop 164	268 Chasing Amy (1997) Drama / Romance; pop 146
9	275 Sense and Sensibility (1995) Drama / Romance; pop 178	475 Trainspotting (1996) Drama; pop 164
10	423 E.T. the Extra-Terrestrial (1982) Children's / Drama / Fantasy / Sci-Fi; pop 167	15 Mr. Holland's Opus (1995) Drama; pop 160

The accompanying recommendations.csv contains all four methods' full lists for both users, plus the controlled PPR lists, with IDs, titles, genres, scores, each movie's training popularity and each list's mean. PPR scores are walk probabilities; their scale is not comparable with MF scores.

5. Discussion

5.1 One decision: test PPR as the main recommender

I recommend PPR as the candidate replacement for the main MF recommendation slot, subject to an online test before rollout. It has the highest observed common-cohort Recall@10 (0.1122 versus MF 0.0547), but its recommendations are more popular on average (244.21 versus 153.34). The evidence supports a retrieval hypothesis, not a claim of better discovery or revenue. The disclosed MF retraining/configuration difference also limits a strict comparison against the historical HW5 model.

User 1 has 217 training ratings, including 131 positive ones; user 2 has 49, including 32 positive ones. Positive-history genre counts include Drama 61/18 and Comedy 39/11 (user 1/user 2); genres overlap. PPR recommends Fargo (Crime/Drama/Thriller) and Contact (Drama/Sci-Fi) for user 1, while user 2 receives Return of the Jedi (Action/Adventure/Romance/Sci-Fi/War) and Dead Man Walking (Drama). These histories provide context but do not prove current intent. MF instead ranks Close Shave, A (Animation/Comedy/Thriller) first for user 1 and Casablanca (Drama/Romance/War) first for user 2; its lower popularity does not by itself demonstrate better novelty.

In the required one-factor demonstration, baseline user 1/alpha 0.85 shares 0 movies with user 2/alpha 0.85 (seed-only change), but 7 with user 1/alpha 0.50 (alpha-only change). For user 1 the mean popularity changes from 245.5 to 228.4 when alpha drops; this is an observation for this pair, not a monotonic novelty rule. At 0.85 the two users' list means are 245.5 and254.9. Lower alpha favors shorter walks but cannot be interpreted as a purchase probability or guaranteed relevance.

Online experiment. Randomize eligible warm users persistently to the existing MF main slot or the same slot served by PPR, keeping position/count/UI fixed and one assignment per user across sessions. Use completed recommended-title views per assigned user as the primary outcome; monitor hide/dislike rate and serving latency as guardrails. Determine sample size and minimum worthwhile effect before running; do not stop when results first look favorable. Offline ratings cannot establish present-session intent, completion, retention or revenue, so no rollout gain is asserted.

5.2 BUILD / ASK WHY / TEST / EXPLAIN / IMPROVE

BUILD: establish graph propagation, four comparators, shared-cohort Recall and full exports. **ASK WHY:** 12 passing self-checks did not explain why both MF lists had identical scores. **TEST:** the earlier export contained 20 MF scores equal to 3.5806026458740234, the float32 training mean; a controlled real-batch diagnostic compared the two regularization normalizations. **EXPLAIN:** the starter added a full-factor L2 sum to mean batch MSE without batch-size normalization, making shrinkage dominate the measured initial gradient. **IMPROVE:** change only the L2 coefficient to reg/actualBatchSize and rerun, selecting the checkpoint using validation rather than test results.

Before repair, MF Recall was 0.0211 and validation RMSE 1.1830 at epoch 1. After repair, the student's WebGL run gives 0.0547 and 0.9878 at epoch 6; each exported MF list now has 10 different scores. This is before/after debugging evidence, not a hyperparameter search for test performance. The five-step micro-diagnostic supports the shrinkage mechanism but alone does not prove long-run collapse.

The earlier alpha sweep also blocked the UI. Batched propagation, asynchronous readback/yield points, caching, progress and cancellation addressed that path; a reproduced tensor-leak fixture changed from 5 / 269 / 533 / 797 tensors to 5 / 5 / 5 / 5. Student checks confirmed cancellation and restart, while automated fixtures checked cache/state isolation. Pre-repair 95.7 s / 0.2 s first/repeat timings concern different work and are not an algorithm speedup claim; the current 154.1 s run is not a repeated benchmark. Next improvement: persist the selected MF weights/configuration and add regression checks for comparator health, alongside bounded browser responsiveness measurements.

6. AI Usage Disclosure

Tools and use. OpenCode 1.19.3, provider/model identifier `opencode/big-pickle`, generated and repaired application code and ran diagnostics. This Codex chat assisted with requirements review, independent export/source checks, manual-check instructions and report preparation; its exact underlying model/version is not established by the saved evidence. Python was used for raw-data/CSV checks and PDF generation; Node.js executed JavaScript checks. TensorFlow.js 4.22.0 and D3 7.9.0 are application libraries, not evidence of personal verification.

What I personally verified. I built the graph and inspected 2,554 nodes and 45,602 positive interactions, ran the browser self-checks and observed 12/0/0, and completed the repaired WebGL evaluation with the displayed four-method results. I hand-calculated 2/19 = 0.105263157 and compared it with the displayed user 1 Recall 0.1053 and the code expression `hit / relevant.size`; they matched. I compared the PPR CSV value 0.11218853219486066, cohort 861, with the page. I also performed the guided demo-slider isolation, repeat-evaluation and Cancel/restart checks on the earlier revision, and exported the corrected run's three evidence files. Earlier UI checks are not presented as newly repeated performance benchmarks after the MF repair.

What I trusted without verification. I did not personally recompute every user's Recall, audit every tensor/kernel or independently validate every bibliographic record. I relied on AI-assisted raw-data checks for all 100 exported recommendation rows, automated memory/cache/completion regressions and citation verification. I did not personally confirm every live progress-counter field, all-user convergence or an online business effect. No such claim is made. The handwritten Recall check and explicit code/output comparison are my own confirmed checks, not agent executions.

Native session. The attached `golankov_alexander_a06_session.json` is the unchanged plaintext native export of the continuous OpenCode session, including prompts, raw tool records and the final L2 repair: 832 messages and 877 tool records at export. Work was resumed after computer shutdowns. The report and its independent checks were prepared in this chat; the OpenCode export is not represented as a record of this separate report-authoring conversation.

References

[1] T. H. Haveliwala, “Topic-Sensitive PageRank,” WWW, 2002. Author paper, section 2.

[2] F. M. Harper and J. A. Konstan, “The MovieLens Datasets: History and Context,” ACM Transactions on Interactive Intelligent Systems, vol. 5, no. 4, article 19, 2015. DOI: 10.1145/2827872. Author manuscript.

[3] GroupLens, “MovieLens 100K Dataset,” released April 1998. Official dataset distribution. Accessed 20 October 2026.

[4] TensorFlow.js, “API reference, version 4.22.0,” tf.matMul, tf.tidy and tf.dispose. Versioned official documentation. Accessed 20 October 2026.

Reproducibility files

GitHub: samgopiter-byte/RecSys-LLMs, branch a06-personalized-pagerank, week6. Run instructions are in readme.md; final machine-readable files are in week6/submission/. The session JSON and this report are delivered as the two course-upload files. Final numbers refer to the 20 October 2026 WebGL run, not a new CPU rerun.